Literature Review & Reference Benchmark

Project Name: Global Energy Shock 2026: Oil, Trade & Economic Exposure Analysis

Program: Digital Egypt Pioneers Initiative (DEPI), Data Analytics Track

1. Review of 3 Authoritative Sources

We reviewed three sources that look at oil shocks from three angles: what is happening in the oil market now, how oil prices affect economies around the world, and how global activity reacts to different kinds of shocks.

Source 1: IEA, Oil Market Report, October 2026

International Energy Agency, Paris. Published 14 October 2026.

https://www.iea.org/reports/oil-market-report-october-2026

Summary

The IEA report describes the oil market during a major Middle East supply disruption. It reports that world oil demand is expected to fall by about 2.6 million barrels per day on average in 2026, and that world supply is expected to fall by about 5.8 million barrels per day. In September 2026, the North Sea Dated price jumped by $26.17 to $116.89 per barrel, and the report notes it was trading near $134 at the time of writing. Brent futures were lower, at around $107, and some physical grades traded far above both. Diesel prices reached record levels. The report says these prices are raising the cost of goods, cutting real incomes, and pushing inflation up. Global stocks are also far below their pre-war level.

Key Takeaway for Our Project

- An oil price spike is not one number. Benchmark prices (Brent, WTI), physical prices and product prices can move by very different amounts. Our dataset holds the Brent and WTI benchmarks, so we label them clearly as *benchmarks*.
- The report shows why our question matters. The shock reaches importers through import bills, freight costs, inflation and incomes. This is the chain our platform follows: **price shock, trade exposure, import bill, GDP exposure**.
- Timing limit: our monthly data stops in May 2026, and our annual data stops in 2025. We cannot observe the September and October 2026 events described by the IEA. Our scenarios are therefore *what-if exposure estimates*, not measurements of this event.

Source 2: Rasmussen & Roitman (IMF), Oil Shocks in a Global Perspective: Are They Really That Bad?

IMF Working Paper WP/11/194, August 2011.

https://www.imf.org/-/media/websites/imf/imported-full-text-pdf/external/pubs/ft/wp/2011/_wp11194.pdf

Summary

The authors study 144 countries from 1970 to 2010 using IMF data on oil prices, GDP, imports and exports. They split countries into oil exporters and three groups of importers (OECD, middle-income and low-income). They find that the impact of higher oil prices on importing economies is usually small: a 25% price

increase typically lowers GDP by about half a percent or less, spread over two to three years. The size of the loss depends mainly on how large oil imports are compared with GDP. A typical importer, with net oil imports of 3 to 4% of GDP, loses about 0.3% of GDP. For importers with oil imports above 5% of GDP, the loss rises to about 1%. The impact is softened because oil exporters spend part of their extra income on imports from other countries, which raises exports and other income for importers. The authors also point out that oil prices often rise in good economic times, when demand is strong.

Key Takeaway for Our Project

- This paper justifies our core metrics. Exposure should be measured **relative to the size of the economy**, which is why we use **Crude Import Burden (% of GDP)** and **Crude Export Dependence (% of GDP)**.
- Importers and exporters must be separated. The paper's exporter group has large net oil exports (26.3% of GDP on average) while importer groups are negative (about -2% to -4%). This supports our **Net Crude Trade** classification and our two-sided scenario (extra cost for importers, extra revenue for exporters).
- Our scenarios show only the direct bill effect. The paper shows that offsetting flows and lagged effects exist, so our scenario results are an exposure measure, not a GDP-loss forecast.
- The authors dropped countries without complete GDP series. This supports our practice of analyzing complete sub-samples instead of filling gaps.

Source 3: Jiménez-Rodríguez, Oil Shocks and Global Economy

Energy Economics, Vol. 115, 2022, Article 106373.

https://www.sciencedirect.com/science/article/pii/S0140988322005023

Summary

The paper studies how global economic activity reacts to three kinds of oil market shocks between 1973 and 2019: an oil supply disruption, a rise in overall demand, and an oil-specific demand increase. It uses a model where the effects are allowed to change over time. The main findings are:

- An unexpected supply disruption lowers global economic activity in every period studied.
- An unexpected rise in overall demand raises global activity in every period.
- An oil-specific demand increase has only a short effect, and mainly in the early 2000s, during the global financial crisis, and in the late 2010s.

The author concludes that investors and policy makers should be careful with oil-specific demand shocks, because their effect depends on when they happen. The paper uses EIA world crude production and a WTI-based real oil price, the same kind of monthly supply and price series that sits in our energy layer.

Key Takeaway for Our Project

- The same price rise can mean different things depending on its cause. A rise caused by lost supply hurts activity, while a rise caused by strong demand can go together with growth. Our price data does not tell us the cause, so **we do not claim that any price move caused a specific economic outcome**.
- This is why the dashboard shows *exposure under a scenario*, and why every result is labelled as a first-order financial estimate.
- Effects change over time, so our year filters and year-over-year views matter. We will not treat one year as typical of all years.

Reference Benchmark Summary

Source	What It Gives Us	How We Use It	Project Element
IEA, Oil Market Report, Oct 2026	Current picture of a price and supply shock, and the gap between benchmark and physical prices	Frame the problem; label Brent/WTI as benchmarks; state our data end dates	Problem statement, price table, scenario labels
IMF WP/11/194	Link between oil imports as % of GDP and size of impact; importer/exporter split	Scale exposure by GDP; classify countries by net trade; keep scenarios two-sided	Net Crude Trade, Import Burden, Export Dependence
Jiménez-Rodríguez (2022)	Shock type and timing change the outcome	Avoid causal claims; use year views; present results as exposure	Dashboard wording, time filters, limitations

2. Methodological Alignment & Lessons Learned

Reading these sources shaped how we designed the data platform.

What We Learned	What We Did in the Project
Exposure has to be compared with the size of the economy (IMF).	We built ratios against GDP: Crude Import Burden, Crude Export Dependence, and Oil Intensity (TWh per $ billion GDP).
Importers and exporters react in opposite ways (IMF).	We classify all 150 countries by Net Crude Trade and calculate both extra import cost and extra export revenue in every scenario.
Price spikes come in different forms, and benchmarks do not show the whole market (IEA, Jiménez-Rodríguez).	We keep Brent and WTI as benchmark series, keep production separate from prices, and avoid any claim of cause and effect.
Monthly market events and annual economic data move at different speeds (IEA, IMF).	We join monthly and annual data through a date dimension and a year dimension, and we average monthly prices to annual values in SQL when comparing with GDP, trade and FX.
Cross-country studies need consistent samples (IMF).	We keep source missing values as they are (V2 rule: missing is not zero), leave ratios blank when an input is missing, and report the number of countries behind every result.
Offsetting flows and lagged effects exist (IMF, Jiménez-Rodríguez).	Scenarios are calculated on observed trade values with volumes fixed. We state that they are exposure estimates, not forecasts or GDP-loss estimates.
Data provenance protects credibility.	Each of the eight tables keeps its source file and grain documented, and the CIF/FOB difference in trade valuation is stated wherever net trade is shown.

Where the literature stops and our dataset starts. The V2 rule on missing values and the CIF/FOB note come from the dataset's own documentation and from UN Comtrade trade conventions. The three sources above support the *principle* of using consistent samples, but they do not set our missing-value rule.

Limits we accept.

- Our monthly series end in May 2026 and our annual series end in 2025, so we cannot observe the autumn 2026 events described by the IEA.
- The IMF thresholds use net oil imports of goods and services. Our measure uses crude imports (HS 2709) from UN Comtrade. We use the paper's thresholds as a *reference*, not as an exact match.

3. Continuous Improvement, Feedback & Evaluation

3.1 Instructor / Evaluator Feedback Log

This section is the official audit log for feedback after the Phase 1 submission. It stays empty until real feedback is received. We will record only feedback that was actually given, with its date and source, and we will not add ratings that have not been confirmed.

#	Date	Reviewer	Feedback Received	Action Taken	Status	Owner
1						
2						
3						

3.2 Proposed Enhancements for Phase 2

1. **Add exposure bands to the What-If simulator.** Group countries by import burden (for example, below 3%, 3 to 4%, 4 to 5%, and above 5% of GDP), inspired by the IMF paper. Let the user pick a price shock and see how many countries fall in each band. Show the results next to the +10%, +25% and +50% scenarios, with a clear note that these are exposure estimates.
2. **Refine the net trade categories.** Go beyond a simple importer/exporter flag. Add a data-coverage flag (complete, partial, no data) and a combined view of import burden and oil intensity, so that countries that look safe only because data is missing are not mistaken for low-risk ones. Some importers also export crude, so we will keep both values visible.
3. **Add careful correlation views.** Test how annual Brent and WTI changes, and the Brent-WTI gap, move together with year-over-year changes in net crude trade by income group. Report the number of observations for every result and label results as associations, not causes.

References

- IEA (2026). *Oil Market Report - October 2026*. IEA, Paris. https://www.iea.org/reports/oil-market-report-october-2026https://www.iea.org/reports/oil-market-report-october-2026
- Rasmussen, T. N., & Roitman, A. (2011). *Oil Shocks in a Global Perspective: Are They Really That Bad?* https://www.imf.org/-/media/websites/imf/imported-full-text-pdf/external/pubs/ft/wp/2011/_wp11194.pdf
- Jiménez-Rodríguez, R. (2022). Oil shocks and global economy. *Energy Economics, 115*, 106373. https://www.sciencedirect.com/science/article/pii/S0140988322005023